

Engineering Review Report

Project NOVA Engineering Assessment Milestone 001

Field	Value
Document ID	NOVA-ERR-001
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Author	Antigravity (Lead Software Engineering Agent)
Reviewer	ChatGPT (Chief Architect)
Approved By	Praveen (Project Founder)
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Dependencies	NOVA-SPEC-001 to NOVA-SPEC-010

Executive Summary

This document serves as the formal Engineering Review Report (Milestone 001) for Project NOVA. Following the stabilization of the repository structure and baseline architectural blueprints, this review analyzes the current documentation landscape against large-scale software industry standards. The report identifies 25 missing documents across 19 categories required before entering the production implementation phase, concluding with a prioritized engineering roadmap.

1. Documentation Quality Assessment

Current State:

- **Foundation & Product (NOVA-SPEC-001, 002):** High quality. Clear personas, scope, and non-functional requirements are established.

- **Architecture (NOVA-SPEC-003, 010):** Excellent conceptual framing. The new 4-Layer Operating Platform model provides a robust mental model. The Communication Framework introduces necessary decoupling.
- **AI Core & Capabilities (NOVA-SPEC-004 to 009):** Strong abstract definitions.

Gap Analysis:

While the *abstract* architecture is sound, the repository entirely lacks *implementation-level* standards. There are no guidelines dictating how code is written, how dependencies are managed, how CI/CD pipelines run, or how security is mathematically enforced. Without these, development will fracture immediately.

2. Missing Engineering Documents

Engineering Manifesto

- **Suggested Name:** Software Engineering Standards & Practices
- **Document ID:** NOVA-ENG-001
- **Purpose:** Define the cultural and technical rules of engineering for the project (Code review checklists, Branching strategies like GitFlow vs Trunk-Based, issue tracking conventions).
- **Dependencies:** NOVA-SPEC-001
- **Priority:** HIGH
- **Recommended Order:** 1

3. Missing Architectural Specifications

Plugin Framework Specification

- **Suggested Name:** Plugin & Extension Framework Specification
- **Document ID:** NOVA-SPEC-013
- **Purpose:** Define how third-party developers build capabilities that plug into the NOVA Kernel without modifying core source code.
- **Dependencies:** NOVA-SPEC-009 (Kernel)
- **Priority:** HIGH
- **Recommended Order:** 4

Memory Architecture Specification

- **Suggested Name:** Persistent Memory Architecture Specification
- **Document ID:** NOVA-SPEC-014
- **Purpose:** Define Vector DB schemas, context chunking, short-term vs long-term storage mechanisms, and PII anonymization.
- **Dependencies:** NOVA-SPEC-004 (AI Core)

- **Priority:** HIGH
- **Recommended Order:** 5

Execution Pipeline Specification

- **Suggested Name:** Execution Pipeline Specification
- **Document ID:** NOVA-SPEC-015
- **Purpose:** Define the step-by-step state machine from the Planner's intent output down to the Capability Router's OS execution, including the Observation Engine's feedback loop.
- **Dependencies:** NOVA-SPEC-009, NOVA-SPEC-004
- **Priority:** MEDIUM
- **Recommended Order:** 6

4. Missing Engineering Standards

Source Control Standards

- **Suggested Name:** Source Control & Branching Strategy
- **Document ID:** NOVA-ENG-002
- **Purpose:** Define commit message formats (e.g. Conventional Commits), PR templates, code owner rules, and merge approval requirements.
- **Dependencies:** None
- **Priority:** HIGH
- **Recommended Order:** 2

5. Missing Capability Specifications

The 4-Layer Operating Platform model introduced new capabilities that have not been specified.

Terminal & Shell Capability

- **Suggested Name:** Terminal Capability Specification
- **Document ID:** NOVA-SPEC-016
- **Purpose:** Define the API for NOVA to execute and read from background CLI environments securely.
- **Dependencies:** NOVA-SPEC-003
- **Priority:** MEDIUM
- **Recommended Order:** 10

File System & Office Capability

- **Suggested Name:** File System & Workspace Capability Specification

- **Document ID:** NOVA-SPEC-017
- **Purpose:** Define how NOVA traverses, reads, edits, and organizes local project directories and Office documents.
- **Dependencies:** NOVA-SPEC-003
- **Priority:** LOW
- **Recommended Order:** 11

6. Missing Security Documentation

Core Security & Permission Policy

- **Suggested Name:** Security Policy & Permissions Specification
- **Document ID:** NOVA-SPEC-011
- **Purpose:** Define the Permission Manager, Tokenized Sandboxing, and Secret Management (API keys). This prevents runaway AI behavior.
- **Dependencies:** NOVA-SPEC-009 (Kernel)
- **Priority:** CRITICAL
- **Recommended Order:** 3

Threat Model

- **Suggested Name:** System Threat Model & Risk Assessment
- **Document ID:** NOVA-SEC-001
- **Purpose:** Document attack vectors (e.g., Prompt Injection, unauthorized capability execution) and mitigations.
- **Dependencies:** NOVA-SPEC-011
- **Priority:** MEDIUM
- **Recommended Order:** 15

7. Missing Testing Documentation

Testing Strategy

- **Suggested Name:** Automated Testing & QA Specification
- **Document ID:** NOVA-ENG-003
- **Purpose:** Define required coverage thresholds, unit test patterns (PyTest), mocking strategies for capabilities, and E2E agent evaluation metrics.
- **Dependencies:** NOVA-ENG-001
- **Priority:** HIGH
- **Recommended Order:** 7

8. Missing Deployment Documentation

Packaging & Release

- **Suggested Name:** Deployment & Packaging Specification
- **Document ID:** NOVA-SPEC-012
- **Purpose:** Define how NOVA is compiled, packaged (Docker vs PyInstaller binaries), versioned, and distributed to end-users.
- **Dependencies:** NOVA-SPEC-003
- **Priority:** MEDIUM
- **Recommended Order:** 17

9. Missing Developer Documentation

Onboarding Guide

- **Suggested Name:** Developer Onboarding & Local Setup Guide
- **Document ID:** NOVA-DEV-001
- **Purpose:** A quick-start guide for new engineers to clone, configure Python virtual environments, install dependencies, and run the system locally.
- **Dependencies:** None
- **Priority:** HIGH
- **Recommended Order:** 8

10. Missing API Documentation

System API Contracts

- **Suggested Name:** API Schema & Interconnect Definitions
- **Document ID:** NOVA-API-001
- **Purpose:** Formal OpenAPI/JSON Schema definitions for the payloads passed across the `IEventBus` and remote WebSockets.
- **Dependencies:** NOVA-SPEC-010
- **Priority:** MEDIUM
- **Recommended Order:** 16

11. Missing Diagrams

- **Network Topology Map:** Missing visual mapping of how a multi-node distributed deployment would route traffic.
- **Database ERD:** Missing Entity-Relationship Diagram for SQLite/Vector memory persistence.
- *Action:* To be embedded within `NOVA-SPEC-014` (Memory) and `NOVA-SPEC-012` (Deployment).

12. Missing ADRs

We are missing Architecture Decision Records for crucial upcoming implementation choices:

- **NOVA-ADR-009:** Concrete Event Bus Technology (asyncio vs Redis).
- **NOVA-ADR-010:** Dependency Injection Framework (Manual vs dependency-injector).
- **NOVA-ADR-011:** UI Framework for Presentation Layer (PyQt vs Electron vs Web UI).
- *Priority:* MEDIUM (Draft concurrently with Milestone 2 implementation).

13. Missing Repository Standards

Structure & Linting

- **Suggested Name:** Repository Linting & Directory Standards
- **Document ID:** NOVA-ENG-004
- **Purpose:** Define pre-commit hooks, Flake8/Black settings, and strict namespace directory policing.
- **Dependencies:** NOVA-ENG-001
- **Priority:** HIGH
- **Recommended Order:** 9

14. Missing Coding Standards

Python Style Guide

- **Suggested Name:** Python Core Coding Guidelines
- **Document ID:** NOVA-ENG-005
- **Purpose:** Mandate async/await paradigms, strict `typing` hints, exception hierarchy, and docstring formats (e.g. Google format).
- **Dependencies:** None
- **Priority:** HIGH
- **Recommended Order:** 12

15. Missing CI/CD Standards

Pipeline Automation

- **Suggested Name:** CI/CD Automation Standards
- **Document ID:** NOVA-ENG-006
- **Purpose:** Define GitHub Actions workflows for PR validation, automated testing, safety scans, and nightly builds.
- **Dependencies:** NOVA-ENG-003 (Testing)
- **Priority:** MEDIUM
- **Recommended Order:** 18

16. Missing Plugin Standards

Plugin Manifest Schema

- **Suggested Name:** Plugin Sandbox & Manifest Standards
- **Document ID:** NOVA-ENG-007
- **Purpose:** Define the exact `manifest.json` schema required to register a 3rd party capability, including required sandbox permissions.
- **Dependencies:** NOVA-SPEC-013
- **Priority:** LOW
- **Recommended Order:** 19

17. Missing Provider Standards

Provider Abstraction Contracts

- **Suggested Name:** Provider Interface Definitions
- **Document ID:** NOVA-ENG-008
- **Purpose:** Low-level specification of the abstract base classes for LLMs, OCR, and TTS/STT engines to guarantee swappability.
- **Dependencies:** NOVA-SPEC-003, NOVA-ADR-005
- **Priority:** HIGH
- **Recommended Order:** 13

18. Missing Interface Specifications

Orchestration IDL

- **Suggested Name:** Core Abstract Interface Definition Language (IDL)
- **Document ID:** NOVA-ENG-009
- **Purpose:** Centralize the raw Python Abstract Base Classes (ABCs) for `IKernel`, `ICapability`, `IEventBus`, `IPlanner` before coding begins.
- **Dependencies:** NOVA-SPEC-009
- **Priority:** CRITICAL
- **Recommended Order:** 14

19. Missing Observability Specifications

Telemetry & Tracing

- **Suggested Name:** Observability & Telemetry Specification
- **Document ID:** NOVA-SPEC-018
- **Purpose:** Define structured JSON logging, distributed correlation IDs, and Prometheus metrics exporting.
- **Dependencies:** NOVA-SPEC-010
- **Priority:** MEDIUM
- **Recommended Order:** 20

20. Recommended Engineering Roadmap

Based on the missing documents above, here is the immediate implementation roadmap for completing the architectural phase before coding begins:

Phase 1: Core Orchestration Finalization

1. **NOVA-ENG-001:** Software Engineering Standards & Practices
2. **NOVA-ENG-002:** Source Control & Branching Strategy
3. **NOVA-SPEC-011:** Security Policy & Permissions Specification (Critical path for Kernel)
4. **NOVA-SPEC-013:** Plugin & Extension Framework Specification
5. **NOVA-SPEC-014:** Persistent Memory Architecture Specification
6. **NOVA-SPEC-015:** Execution Pipeline Specification

Phase 2: Engineering Guardrails

7. **NOVA-ENG-003:** Automated Testing & QA Specification
8. **NOVA-DEV-001:** Developer Onboarding & Local Setup Guide

9. **NOVA-ENG-004:** Repository Linting & Directory Standards
10. **NOVA-SPEC-016:** Terminal Capability Specification
11. **NOVA-SPEC-017:** File System Capability Specification

Phase 3: Pre-Implementation Standards

12. **NOVA-ENG-005:** Python Core Coding Guidelines
13. **NOVA-ENG-008:** Provider Interface Definitions
14. **NOVA-ENG-009:** Core Abstract Interface Definition Language (IDL)
15. **NOVA-SEC-001:** System Threat Model & Risk Assessment
16. **NOVA-API-001:** API Schema & Interconnect Definitions

Phase 4: Day-2 Operations Planning

17. **NOVA-SPEC-012:** Deployment & Packaging Specification
18. **NOVA-ENG-006:** CI/CD Automation Standards
19. **NOVA-ENG-007:** Plugin Sandbox & Manifest Standards
20. **NOVA-SPEC-018:** Observability & Telemetry Specification